Total No. of Questions : 4]

SEAT No. :

PE-541

[Total No. of Pages : 2

[6578]-14

S.E. (Electronics/E & TC Engineering) (Insem.)

ELECTRICAL CIRCUITS

(2019 Pattern) (Semester - III) (204183)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
2) *Figures to the right indicate full marks.*
3) *Neat diagrams and wave forms must be drawn wherever necessary.*
4) *Assume suitable data if necessary.*
5) *Use of nonprogrammable calculator is allowed.*

Q1) a) Find the current through the 2 Ω resistor in the network of Fig. 1 **[5]**

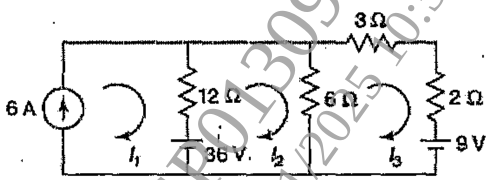

Fig.1

b) Using Norton's theorem, find the current through the 10 Ω resistor in the network shown in Fig.2. **[5]**

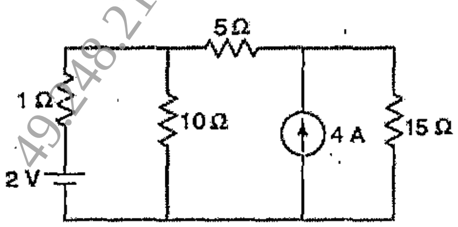

Fig.2

c) Find the power delivered by the 50 V source in the network Fig.3. **[5]**

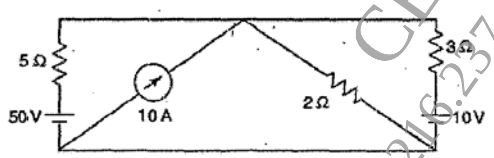

Fig.3

P.T.O.

OR

Q2) a) Using super node analysis, determine the current in the 5 Ω resistor for the network shown in Fig.4. **[5]**

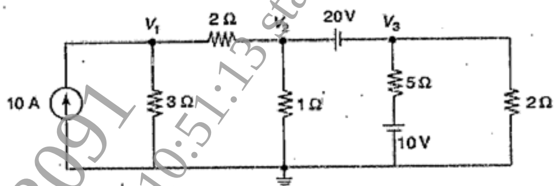

Fig.4

b) Using Thevenin's theorem, find the current through the 1 Ω resistor in Fig.5. **[5]**

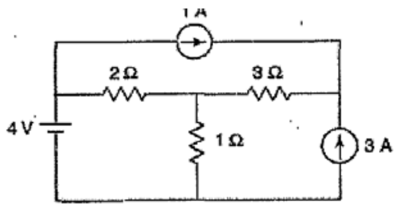

Fig.5

c) State Kirchhoff's voltage and current laws. **[5]**

Q3) a) A series R-L circuit is driven by DC Voltage source V. A voltage source is connected in series circuit through a switch K which is closed at t=0. Find expression for current through inductor for t > 0. Sketch the variation of i(t) against time. **[7]**

b) In the given network of fig.6, the switch is closed at t = 0. With zero current in the inductor, find i, di/dt and di^2/dt^2 at t = 0^+. **[8]**

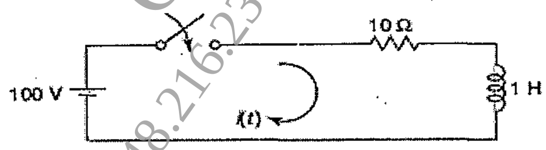

fig.6

OR

Q4) a) Derive the equation for the complementary solution for Voltage across capacitor for driven R-C Circuit. Draw natural Response of the Circuit for various values of t. **[7]**

b) In the network of Fig.7 the switch is closed at t = 0. With the capacitor uncharged, find value for i, di/dt and di^2/dt^2 at t = 0^+. **[8]**

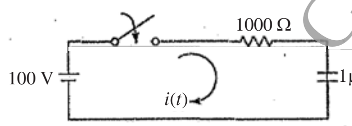

fig.7

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